■ Disease Detectives

HUMANS VS. BACTERIA · ACTIVITY 3 · EXTRA CREDIT

■ Due Tuesday · Worth up to 25 points · Can only HELP your grade ■

■ The Big Idea

Here's something wild: in 1854, a doctor named John Snow stopped a cholera outbreak in London — not with medicine, but with a **map**. He plotted every death on paper, noticed they all clustered around one water pump on Broad Street, and had the handle removed. The outbreak stopped almost immediately. No antibiotics. No vaccine. Just data and a sharp eye.

That's exactly what you did in the simulation — you watched disease spread, changed variables, and looked for patterns. John Snow did that in real life, except people were actually dying.

Fast forward to right now: NASA just discovered a bacterium hiding in their supposedly sterile spacecraft clean rooms. It survives by **playing dead** — going completely dormant so scientists can't detect it, then waking back up when conditions improve. Sound familiar? That's the same idea as the immune gene variants that helped some people survive the Black Death — **variation in how organisms respond to stress determines who lives and who doesn't**.

Bacteria have been outsmarting us for a long time. Your job is to find out how we're fighting back.

■ Here's how it all connects:

- ■■ **John Snow** → used data and a map to find the source of disease
- ■■ **The simulation** → showed you how crowding, mixing, and variation drive spread
- ■■ **The Black Death** → variation in immune genes determined who survived
- ■ **NASA bacteria** → bacteria using their own variation (dormancy) to survive our best defenses

It's all one story: bacteria are sneaky, and humans have to get smarter to keep up.

1 Start With John Snow

No new research needed — use your class materials

Before you look anything up, go back to the **cholera maps from class**. These two questions are your foundation — everything else builds on them.

Answer both of these in Slide 2 of your presentation:

Question A: Which water pump had the contaminated water? How can you tell just by looking at the map? (Hint: look at where the deaths cluster.)

Question B: Why did some people who lived *far* from that pump still get sick? Use what you learned about **mixing and travel** from the simulation to explain it.

2 Find a Current Outbreak

This is where the research happens

Find a disease outbreak happening **right now or within the last 2 years**. Pick one that interests you:

	Disease	Why it's interesting
A	Bird Flu (H5N1)	Spreading in ways scientists are actively watching — real data available now
B	Mpox (Monkeypox)	Recent global outbreak with clear crowding + mixing patterns
C	Measles Outbreak	Happening in multiple countries — connects directly to vaccination
D	**Wild Card**	**Make the case that NASA's 'zombie bacteria' (Tersicoccus phoenicis) could become a future public health problem. Use the NatGeo article + 1 more source.**

Source rules: Use at least **2 sources** — CDC (cdc.gov), WHO (who.int), or reputable news (NYT, BBC, National Geographic). No Wikipedia as your main source.

3 Build Your Slideshow

Using Google Slides

How to set up Google Slides:

1. Go to **slides.google.com**
2. Click the big **+** button (Blank presentation)
3. Click the title and rename it: **Disease Detectives — [Your Name]**
4. When done, click **Share** (top right) → **Anyone with the link can view** → copy and paste the link into Canvas.

■ What goes on each slide:

1 Title Slide

- Your name, your disease, and today's date
- Add a photo or image related to your disease — something visual, not just words

2 The John Snow Connection

- Answer both questions from Step 1 here
- Include at least one of the cholera maps from class (screenshot it and insert it)
- Label the contaminated pump on the map if you can

3 What Is This Disease?

- What pathogen causes it? (bacteria, virus, etc.)
- What are the symptoms?
- How does it spread from person to person?
- Keep it to bullet points — no paragraphs

4 Where Is It Happening?

- Find a map showing where the outbreak is occurring and insert it
- Write 2–3 sentences describing where it's spreading and how fast

5 The Numbers

- How many people have been infected? How many deaths?
- If you can find a graph or chart showing spread over time, include it
- If not, make a simple table with the numbers you found

6 Connect to Activity 3 — The Simulation ■

- This is your most important analysis slide — spend the most time here
- Which simulation factors show up? Crowding? Mixing/travel? Poor hygiene? Variation?
- Give a specific example for at least 2 factors
- Example: "Bird flu spreads via migration routes — just like high mixing in the sim"

7 Connect to the NASA Bacteria Story

- Does your disease do anything 'sneaky' to survive or spread?
- Does variation play a role in who gets sicker or who recovers?
- Think: antibiotic resistance, immune variation, mild vs. severe cases

8 What's Being Done to Stop It?

- Vaccines, antibiotics, quarantine, public health campaigns?
- What do we have now that people in 1347 didn't?
- What would John Snow think about how we're handling it?

9 Sources

- List every source you used
- Format: Website Name — Article Title — URL
- You need at least 2 sources

■ Design tips:

■■ Every slide needs at least one image, map, or chart — no walls of text

■ Use bullet points, not paragraphs

■ Font size at least 20 for body text — make it readable

■ Slide 6 is your most important slide — give it your best work

4 Present to the Class

3–5 minutes — you've got this!

Walk us through your slides in a future session. You don't need to memorize anything — just talk through what you found like you're explaining it to a friend. You already know this material. This is just

showing your work.

■ How You Earn the 25 Points

Category	What we're looking for	Pts
John Snow connection (Slide 2)	Both questions answered, map included, pump identified	6
Current outbreak research (Slides 3–5)	Disease explained clearly, map or image, real numbers, 2+ sources	7
Activity 3 connection (Slide 6)	At least 2 simulation factors with specific real-world examples	5
NASA bacteria / variation (Slide 7)	A real connection made between your disease and variation or dormancy	4
Presentation (Step 4)	Showed up and talked through it — doesn't have to be perfect	3
TOTAL		**25**

■ **Remember:** This assignment can **only help** your grade. You cannot lose points on Activity 3 by trying this. The 25 points go on top of whatever you already have. If you try and it's not your best work, you just don't get the extra credit — nothing is taken away. So go for it! ■